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Studierichting: Informatica
Oefeningenreeks 1B nr. 37.9

$$\sqrt{-1 - 2\sqrt{2}i}$$

$$-1 - 2\sqrt{2}i = (x + yi)^2$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ 2xy = -2\sqrt{2} \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ xy = -\sqrt{2} \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 = -1 \\ x^2 y^2 = (-\sqrt{2})^2 = 2 \end{cases}$$

x^2 en y^2 zijn oplossingen van:

$$\lambda^2 - \lambda - 2 = 0$$

$$\Delta = 1 + 4 \cdot 2 = 9$$

$$\lambda_{1,2} = \frac{1 \pm \sqrt{9}}{2}$$

$$\lambda_1 = \frac{4}{2} = 2 \text{ en } \lambda_2 = -1$$

$$x = \sqrt{2} \text{ en } y = 1$$

Oplossing: $\sqrt{2} - i$ of $-\sqrt{2} + i$

References